

INTRODUCTION TO APPLIED NUMERICAL METHODS

Linear Algebra Background & Application

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LINEAR ALGEBRAIC EQUATIONS

What is linear equations?

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\&\vdots \\&\vdots \\a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n\end{aligned}$$

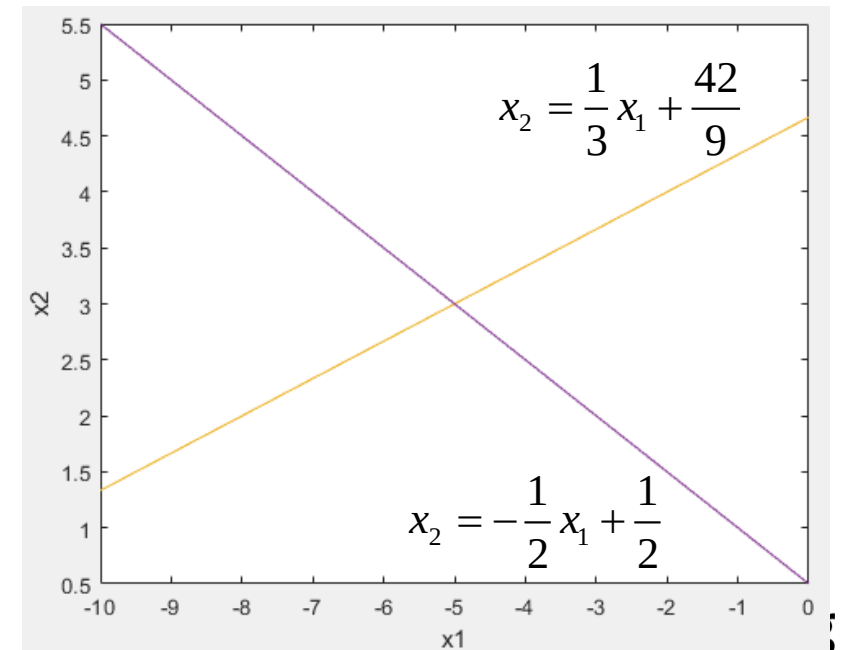
a - constant coefficients;
 b -constants;
 x - unknowns to be solved

Mathematical Meanings

$$\begin{array}{l}3x_1 - 9x_2 = -42 \\2x_1 + 4x_2 = 2\end{array} \Rightarrow \begin{array}{l}9x_2 = 3x_1 + 42 \\4x_2 = -2x_1 + 2\end{array} \Rightarrow \begin{array}{l}x_2 = \frac{1}{3}x_1 + \frac{42}{9} \\x_2 = -\frac{1}{2}x_1 + \frac{1}{2}\end{array}$$

Solutions to x 's are the intersection of lines (2D), planes (3D), and hyper-planes

x 's need to satisfy all equations!



MATRIX NOTATION

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2m} \\ \cdot & \cdot & & & \cdot \\ \cdot & \cdot & & & \cdot \\ \cdot & \cdot & & & \cdot \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nm} \end{bmatrix}$$

Column 3

Row 2

Numerical Methods for Engineers, 7th Ed.
(S.C.Chapra and R.P.Canale)

- A matrix includes a rectangular array of elements represented by a single symbol
- $[A]$ (or A) is the shorthand notation for the matrix
- a_{ij} is the individual element of the matrix forming matrix
- First subscript i : the number of the row; 2nd subscript j : the number of columns
- 3 n rows and m columns: dimension (nXm)

MATRIX NOTATION

Examples

$$[A] = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

- $a_{11} = ??$
- $a_{21} = ??$
- $a_{43} = ??$
- First row??
- Third column??

Answer:

- $a_{11} = 1$
- $a_{21} = 5$
- $a_{43} = 15$
- $[1 \ 2 \ 3 \ 4]$

$$\begin{bmatrix} 3 \\ 7 \\ 11 \\ 15 \end{bmatrix}$$

VECTOR NOTATION

Matrices with row dimension $n = 1$

$$[C] = [c_1 \quad c_2 \quad \dots \quad c_m]$$

Drop the first subscript

Row Vectors $[C]$

Example

$$[C] = [1 \quad 2 \quad 3 \quad 4]$$

Matrices with column dimension $m = 1$

$$[B] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix}$$

Drop the 2nd subscript

Column Vectors $\{B\}$

Example:

$$[B] = \begin{bmatrix} 3 \\ 7 \\ 11 \\ 15 \end{bmatrix}$$

SPECIAL TYPES OF SQUARE MATRICES

Symmetric Matrix

$$a_{ij} = a_{ji}$$

$$[A] = \begin{bmatrix} 5 & 1 & 2 \\ 1 & 3 & 7 \\ 2 & 7 & 8 \end{bmatrix}$$

Principal or main diagonal

Diagonal Matrix

$$[A] = \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & a_{33} \\ & & & a_{44} \end{bmatrix}$$

$$[A] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Identity Matrix

$$[I] = \begin{bmatrix} 1 & & 0 \\ & 1 & \\ 0 & & 1 \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

SPECIAL TYPES OF SQUARE MATRICES

Upper Triangular Matrix

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ & a_{22} & a_{23} & a_{24} \\ & & a_{33} & a_{34} \\ & & & a_{44} \end{bmatrix}$$

Lower Triangular Matrix

$$[A] = \begin{bmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ a_{31} & a_{32} & a_{33} & \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

Banded Matrix

$$[A] = \begin{bmatrix} & a_{12} & & \\ a_{21} & a_{22} & a_{23} & \\ & & a_{32} & a_{33} & a_{34} \\ & & & a_{43} & a_{44} \end{bmatrix}$$

Bandwidth = 3
Tridiagonal matrix

$$[A] = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & 11 & 12 \\ 0 & 0 & 0 & 16 \end{bmatrix}$$

$$[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 5 & 6 & 0 & 0 \\ 9 & 10 & 11 & 0 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

$$[A] = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 5 & 6 & 7 & 0 \\ 0 & 10 & 11 & 12 \\ 0 & 0 & 15 & 16 \end{bmatrix}$$

MATRIX OPERATING RULES

$$[A] = [B]$$

- $a_{ij} = b_{ij}$ for all i and j

Matrix Addition and Subtraction

$$[C] = [A] + [B] \quad \Rightarrow \quad c_{ij} = a_{ij} + b_{ij}$$

$$[D] = [E] - [F] \quad \Rightarrow \quad d_{ij} = e_{ij} - f_{ij}$$

- Commutative

$$[A] + [B] = [B] + [A]$$

- Associative

$$([A] + [B]) + [C] = [A] + ([B] + [C])$$

**All matrices have
the same dimension
 $n \times m$!**

$$[A] = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix} \quad [B] = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \Rightarrow \quad [A] + [B] = \begin{bmatrix} 3 & 6 & 9 \\ 12 & 15 & 18 \\ 21 & 24 & 27 \end{bmatrix} \quad [A] - [B] = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

MATRIX OPERATING RULES (MULTIPLICATION)

Matrix Multiplication with a Scalar

$$[D] = g[A] = \begin{bmatrix} ga_{11} & ga_{12} & \cdots & ga_{1m} \\ ga_{21} & ga_{22} & \cdots & ga_{2m} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ ga_{n1} & ga_{n2} & \cdots & ga_{nm} \end{bmatrix}$$

$$[A] = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix} \Rightarrow 5[A] = \begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

MATRIX OPERATING RULES (MULTIPLICATION)

Product of Two Matrices

$$[C] = [A] \cdot [B] \quad \Rightarrow \quad c_{ij} = \sum_{k=1}^m a_{ik} b_{kj}$$

m = the column dimension of $[A]$ and the row dimension of $[B]$

$$[A]_{n \times m} \quad [B]_{m \times l} = [C]_{n \times l}$$

Interior dimensions are equal; multiplication is possible

Exterior dimensions define the dimensions of the result

the number of columns of the first matrix must be equal to the number of rows (m) in the second matrix

MATRIX OPERATING RULES (MULTIPLICATION)

$$[A] = \begin{bmatrix} 3 & 1 \\ 8 & 6 \\ 0 & 4 \end{bmatrix} \quad [B] = \begin{bmatrix} 5 & 9 \\ 7 & 2 \end{bmatrix} \quad [C] = [A][B] = \begin{bmatrix} 3 & 1 \\ 8 & 6 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 5 & 9 \\ 7 & 2 \end{bmatrix} ???$$

Raise [Y] to align the appropriate rows and columns

$$[A] \rightarrow \begin{bmatrix} 3 & 1 \\ 8 & 6 \\ 0 & 4 \end{bmatrix} \quad \begin{bmatrix} 5 & 9 \\ 7 & 2 \end{bmatrix} \leftarrow [B]$$

$c_{11} \quad c_{21} \quad c_{22}$

$$\leftarrow [C]$$

$$\begin{bmatrix} 3 & 1 \\ 8 & 6 \\ 0 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \times 5 + 1 \times 7 = 22 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 9 \\ 7 & 2 \end{bmatrix} \downarrow$$

$$\begin{bmatrix} 3 & 1 \\ 8 & 6 \\ 0 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 22 \\ 8 \times 5 + 6 \times 7 = 82 \\ \end{bmatrix} \quad c_{21}$$

$$[C] = \begin{bmatrix} 22 & 29 \\ 82 & 84 \\ 28 & 8 \end{bmatrix}$$

the number of columns of the first matrix must be equal to the number of rows in the second matrix

PROPERTIES OF MATRIX MULTIPLICATION

- Multiplication is **not generally commutative**

$$[A][B] \neq [B][A]$$

generally $n \neq l$



$$[A]_{n \times m} [B]_{m \times l} = [C]_{n \times l}$$

Interior dimensions are equal; multiplication is possible

Exterior dimensions define the dimensions of the result

$$[A] = \begin{bmatrix} 3 & 1 \\ 8 & 6 \\ 0 & 4 \end{bmatrix} \quad [B] = \begin{bmatrix} 5 & 9 \\ 7 & 2 \end{bmatrix}$$

$$[C] = [A][B] = \begin{bmatrix} 22 & 29 \\ 82 & 84 \\ 28 & 8 \end{bmatrix}$$

$$[B][A]$$

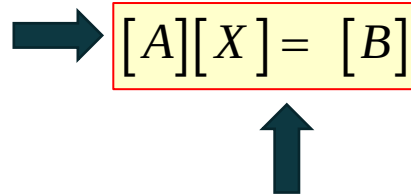


The column number of [B] (2) and the row number of [A] (3) are NOT equal!

REPRESENTING LINEAR ALGEBRAIC EQUATIONS IN MATRIX FORM

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Linear algebraic equations


$$[A][X] = [B]$$

Matrix Form

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

[A] is the n by n square matrix of coefficients

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

[X] is the n by 1 column vector of unknowns

$$[B] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix}$$

[B] is the n by 1 column vector of constants

REPRESENTING LINEAR ALGEBRAIC EQUATIONS IN MATRIX FORM

Prove

$$x_1 + 2x_2 + 3x_3 = 4$$

$$3x_1 + 2x_2 + x_3 = 5$$

$$4x_1 + x_2 + 3x_3 = 6$$



$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 4 \\ 5 \\ 6 \end{Bmatrix}$$

[A] [X] [B]

Step 1: Matrix multiplication using the matrix form

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

[X] [B]

$$\begin{aligned} b_1 &= x_1 + 2x_2 + 3x_3 \\ b_2 &= 3x_1 + 2x_2 + x_3 \\ b_3 &= 4x_1 + x_2 + 3x_3 \end{aligned}$$

$$[B] = \begin{Bmatrix} 4 \\ 5 \\ 6 \end{Bmatrix}$$

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 4 \\ 3x_1 + 2x_2 + x_3 &= 5 \\ 4x_1 + x_2 + 3x_3 &= 6 \end{aligned}$$

Step 2: Because

SPECIAL MATRIX: SQUARE MATRIX

Matrices where $n = m$ are called *square matrices*

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix}$$

$$x_1 + 2x_2 + 3x_3 = 4$$

$$3x_1 + 2x_2 + x_3 = 5$$

$$4x_1 + x_2 + 3x_3 = 6$$



$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 4 \\ 5 \\ 6 \end{Bmatrix}$$

- Number of unknowns and the number of equations are equal for unique solution
- Important for solving the set of linear equations

MATRIX TRANSPOSE

- Matrix Transpose
 - Transforming its rows into its columns and its columns into rows
 - Swap a_{ij} and a_{ji}

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \xrightarrow{\text{Transpose: T}} [A]^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{bmatrix}$$

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$[A]^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$[B] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} \xrightarrow{\quad} [B]^T = [b_1 \quad b_2 \quad \dots \quad b_{n-1} \quad b_n]$$

$$[B] = \begin{bmatrix} 3 \\ 7 \\ 11 \\ 15 \end{bmatrix} \xrightarrow{\quad} [B]^T = [3 \quad 7 \quad 11 \quad 15]$$

MATRIX AUGMENTATION

- Matrix Augmentation
 - Attaching a column (or columns) to the original matrix

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \xrightarrow{\text{Augmentation}} \left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 1 \end{array} \right]$$

Augment $[A]$ with $[B]$: $[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{bmatrix}$

- Very useful when a set of identical operations needed to be performed on two matrices (or vectors)